PROGRAM NO. 5

Implementing Model-View-Controller (MVC) Architecture in Java

Problem Description:

Student must implement the MVC pattern for student management system where student details (name, roll number) are stored in a model, displayed through a view, and controlled via a controller. The system should follow the Model-View-Controller (MVC) architecture to ensure separation of concerns, making the code maintainable and scalable.

Method

1. Model (Student): Represents the data structure for storing student details. It contains getter and setter methods for data manipulation.
2. View (StudentView): Responsible for displaying student information to the user. It takes data from the model and presents it.
3. Controller (StudentController): Acts as a mediator between the model and the view. It handles user input, updates the model, and refreshes the view accordingly.
4. Client (MVCPatternDemo): Initializes the model, view, and controller, demonstrating how they interact.

CODE:

```
public class MVCPatternDemo {


  // ===== Model =====

  static class Student {

    private String name;

    private String rollNo;


    public String getName() {

      return name;

    }


    public void setName(String name) {

      this.name = name;
```

```
    }

    public String getRollNo() {
        return rollNo;
    }

    public void setRollNo(String rollNo) {
        this.rollNo = rollNo;
    }
}

// ===== View =====
static class StudentView {
    public void printStudentDetails(String name, String rollNo) {
        System.out.println("Student Details:");
        System.out.println("Name: " + name);
        System.out.println("Roll No: " + rollNo);
    }
}

// ===== Controller =====
static class StudentController {
    private Student model;
    private StudentView view;

    public StudentController(Student model, StudentView view) {
        this.model = model;
        this.view = view;
    }

    public void setStudentName(String name) {
```

```
        model.setName(name);
    }

    public void setStudentRollNo(String rollNo) {
        model.setRollNo(rollNo);
    }

    public String getStudentName() {
        return model.getName();
    }

    public String getStudentRollNo() {
        return model.getRollNo();
    }

    public void updateView() {
        view.printStudentDetails(model.getName(), model.getRollNo());
    }
}

// ===== Main Method =====
public static void main(String[] args) {

    // Create Model
    Student model = new Student();
    model.setName("Abhi");
    model.setRollNo("101");

    // Create View
    StudentView view = new StudentView();
```

```
        // Create Controller
        StudentController controller = new StudentController(model, view);

        // Display initial data
        controller.updateView();

        System.out.println("\nUpdating Student Details...\n");

        // Update data using controller
        controller.setStudentName("Rahul");
        controller.setStudentRollNo("102");

        // Display updated data
        controller.updateView();
    }
}
```

OUTPUT:

Student Details:

Name: Abhi

Roll No: 101

Updating Student Details...

Student Details:

Name: Rahul

Roll No: 102